

A Supersonic Interface Obstruction to Classical Evolution

A partial answer to arXiv:2302.03344

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22 July 2026

Status. Candidate partial result, likely valid, pending human review. The theorem is a complete counterexample to sufficiency of the source’s Kato-stability hypotheses for classical evolution-family generation. It is not a full characterization of the open question.

1 Source question

Kilian, Maschke, Mironchenko, and Wirth study a non-autonomous port-Hamiltonian family associated with two systems separated by a moving interface [1]. In the abstract, and again in Section VI, they state that the general conditions under which the family generates an evolution family remain open. Their assumptions require the path $l: [0, \tau] \rightarrow (a, b)$ to be continuously differentiable, but place no bound on the interface speed.

The relevant source statement is reproduced below from PDF page 1.

A case study of port-Hamiltonian systems with a moving interface

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arXiv:2302.03344v2 [math.AP] 12 May 2023

Abstract—We model two systems of two conservation laws defined on complementary spatial intervals and coupled by a moving interface as a single non-autonomous port-Hamiltonian system, and provide sufficient conditions for its Kato-stability. An example shows that these conditions are quite restrictive. The more general question under which conditions an evolution family is generated remains open.

Keywords: port-Hamiltonian systems, strongly continuous semigroups, linear systems, distributed parameter systems, moving interface, Kato-stability.

I. INTRODUCTION

The port-Hamiltonian framework [1], [2], [3] is a systematic theory for modeling, analysis and control of *physical systems* belonging to mechanical, electric, hydraulic, thermal, and other domains. It rests on two eponymous concepts: the system's *Hamiltonian* (that is, its total energy) stemming from mechanics [4], and the *port-based modeling approach* [1], which is based on the power-conserving interconnection of various system components by means of ports.

Following the semigroup approach [5], port-Hamiltonian distributed parameter systems can be represented as abstract Cauchy problems of the form

$$\begin{aligned}\dot{x}(t) &= A_Q x(t), \quad t > 0, \\ x(0) &= x_0 \in X,\end{aligned}$$

where $A_Q: D(A_Q) \subset X \rightarrow X$ is a linear unbounded operator on a Hilbert space X (the so-called *port-Hamiltonian operator*) associated with the port-Hamiltonian system. The port-Hamiltonian operator stores information concerning the interdomain coupling of the multi-physics system, as well as the mechanical and physical properties. Frequently, $A_Q = \mathcal{J}(\mathcal{Q})$, where $\mathcal{J}$ is a formally skew-symmetric matrix differential operator modeling the interdomain coupling, and $\mathcal{Q}$ is a coercive and bounded multiplication operator representing the physical properties [6, Section 3], [1, pp. 226–228]. As discussed in [7], [8], a significant research effort was devoted to the analysis of port-Hamiltonian systems connected via interfaces. Such systems appear, e.g., due to the distinct physical properties on

the respective subdomains. To cover couplings of systems with a non-fixed boundary between them in port-Hamiltonian terms, in [9] the authors proposed the port-Hamiltonian formulation for two systems that are coupled by a moving interface. However, [9] was concentrated on the modeling, and the well-posedness of the obtained model has not been investigated in [9].

In [7], we proposed a mathematically rigorous approach for the study of port-Hamiltonian systems with an interface as abstract evolution equations. Using the framework of [2], easy-to-check criteria for boundary and interface conditions that guarantee that the system with a fixed interface induces a contraction semigroup were given in [7]. Under somewhat stronger conditions, exponential stability of such a system was established in [7]. These findings have been applied in [7] to two acoustic waveguides coupled by an interface consisting of some membrane and in [8] to the interface coupling of two lossless transmission lines.

Contribution. In [7] only systems with a stationary interface have been considered. In this note, we show that the approach advocated in [7] is applicable also to systems with a moving interface. In Section II we consider a port-Hamiltonian formulation of the system with an interface. We introduce the formally skew-symmetric differential operator associated with the aggregate system. Furthermore, we define the boundary and interface port variables in case of a fixed interface position, and state an energy balance equation for this class of systems. In Section III we shall define a family of port-Hamiltonian operators that keeps track of the position of the moving interface, and analyze the resulting time-variant evolution problem associated with this family. We present sufficient conditions for the Kato-stability of this family in the sense of Pazy [10, Section 5.2]. This is a key property for the further analysis of the system formulation suggested in [9] and [8].

Notation. Throughout the paper, X is a Hilbert space and $\mathcal{L}(X)$ is the space of linear bounded operators on X . We denote by $\mathcal{C}([a, b], X)$ and $\mathcal{C}^k([a, b], X)$ the vector spaces of X -valued, continuous and k -times continuously differentiable functions $f: [a, b] \rightarrow X$. Also, $L^2([a, b], \mathbb{R}^n)$ denotes the vector space of $\mathbb{R}^n$ -valued square-integrable functions on the interval $[a, b]$, and $L^\infty([a, b], \mathbb{R}^n)$ is the vector space of $\mathbb{R}^n$ -valued essentially bounded functions on $[a, b]$. Moreover, $H^1([a, b], \mathbb{R}^n)$ denotes the Sobolev space of order 1 on $[a, b]$. For real, square, symmetric matrices A, B the expression $A \leq B$ means that $B - A$ is symmetric positive semi-definite. For functions $y: [a, b] \setminus \{l\} \rightarrow \mathbb{R}^n$ the notations $y(l^+), y(l^-)$ denote the right (resp. left) limit of y at l , assuming existence.

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The source proves Kato stability under assumptions (A1)–(A2). The result below shows that these assumptions cannot by themselves yield a classical evolution family on the natural operator domains. A subcharacteristic speed condition is already necessary in the constant unit medium.

2 Definitions and normalization

Let $X = L^2([-2, 2], \mathbb{R}^2)$, let

$$P_1 = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}, \quad \mathcal{Q}^- = \mathcal{Q}^+ = I_2,$$

and take the moving interface $l(t) = 2t$ for $0 \leq t \leq \tau$, with $0 < \tau < 1/6$. Away from $z = l(t)$ the source equation becomes

$$x_t = P_1 x_z. \quad (1)$$

Set $r = 0$ in the source interface relation. Since its interface flow is the common trace of the second component, every classical solution must satisfy

$$x_2(t, l(t)^-) = x_2(t, l(t)^+) = 0. \quad (2)$$

At the outer endpoints impose $x_2(t, -2) = x_2(t, 2) = 0$. In the source's boundary variables this is obtained from

$$W_B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad W_B \begin{bmatrix} f_\partial \\ e_\partial \end{bmatrix} = 0.$$

Indeed, the two selected equations are $f_{\partial,1} = 0$ and $e_{\partial,2} = 0$, which are equivalent to the two endpoint conditions. For

$$\Sigma = \begin{bmatrix} 0 & I_2 \\ I_2 & 0 \end{bmatrix}$$

we have $\text{rank } W_B = 2$ and $W_B \Sigma W_B^\top = 0$. Thus (A2) holds. The unit Hamiltonian densities satisfy (A1). Consequently the source's Kato-stability corollary applies to this family.

By a *classical evolution family on the natural domains* we mean an evolution family $U(t, s)$ for which every $x_s \in D(A(s))$ gives, at least on a right neighborhood of s , a classical solution $x(t) = U(t, s)x_s$ of $\dot{x}(t) = A(t)x(t)$ satisfying the source trace conditions. This is the notion ruled out below.

3 Proof intuition

The matrix P_1 has characteristic speeds $+1$ and -1 . The interface moves to the right with speed 2, so it overtakes both characteristic families on its right. Therefore both traces at the right side of the interface are determined by the initial datum; the interface cannot prescribe a relation between them. We choose a smooth datum for which those two outgoing traces immediately differ. The source condition nevertheless requires them to be equal, giving an exact contradiction.

4 The counterexample

Theorem 1 (Supersonic obstruction). *For the family just defined, every frozen operator satisfies the source hypotheses (A1)–(A2), and the family is Kato-stable by the source result. Nevertheless, there exists $x_0 \in C_c^\infty((-2, 2), \mathbb{R}^2) \cap D(A(0))$ for which*

$$\dot{x}(t) = A(t)x(t), \quad x(0) = x_0,$$

has no classical solution on any interval $[0, \varepsilon]$ with $\varepsilon > 0$. Hence the family cannot generate a classical evolution family on its natural domains.

Proof. Choose $\eta \in C_c^\infty((-1, 1))$ with $\eta = 1$ on $[-1/2, 1/2]$, and put

$$x_0(z) = \frac{z\eta(z)}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (3)$$

This datum is smooth and compactly supported, satisfies both outer boundary conditions, and has $x_{0,2}(0) = 0$. It therefore belongs to $D(A(0))$.

Introduce the characteristic variables

$$u = x_1 + x_2, \quad v = x_1 - x_2.$$

Equation (1) diagonalizes to

$$u_t + u_z = 0, \quad v_t - v_z = 0. \quad (4)$$

For the datum (3),

$$u_0(z) = z\eta(z), \quad v_0(z) = 0. \quad (5)$$

Assume for contradiction that a classical solution exists. Fix sufficiently small $t > 0$ and consider the right trace at $(t, l(t)) = (t, 2t)$. The backward u -characteristic through this point is

$$z_u(s) = 2t - (t - s) = t + s,$$

and for $0 \leq s < t$ it satisfies

$$z_u(s) - l(s) = t - s > 0.$$

Thus it stays strictly in the right subsystem and reaches time zero at $z_u(0) = t$. Similarly, the backward v -characteristic is

$$z_v(s) = 2t + (t - s) = 3t - s,$$

with

$$z_v(s) - l(s) = 3(t - s) > 0,$$

and it reaches time zero at $z_v(0) = 3t$. Neither trace can be affected by the moving interface or by an outer boundary. The standard characteristic identity for (4), valid also for the H^1 traces of a classical solution, therefore gives

$$u(t, 2t^+) = u_0(t), \quad v(t, 2t^+) = v_0(3t). \quad (6)$$

For $0 < t < 1/6$, both t and $3t$ lie in $(-1/2, 1/2)$, so (5)–(6) yield

$$u(t, 2t^+) = t, \quad v(t, 2t^+) = 0.$$

Consequently

$$x_2(t, 2t^+) = \frac{u(t, 2t^+) - v(t, 2t^+)}{2} = \frac{t}{2} \neq 0,$$

contradicting the source interface condition (2). No classical solution exists for this natural-domain datum. $\square$

5 What the theorem establishes

The example meets the source's structural and boundary hypotheses exactly; in particular it is not the non-Kato-stable example already given in the source. It proves that Kato stability of the frozen family does not prevent a moving interface from overdetermining outgoing characteristic traces. In the normalized unit medium, an all-data classical generation theorem must at least exclude rightward speeds $l' > 1$. Reflecting the construction excludes leftward speeds $l' < -1$ as well.

This packet does not prove sufficiency when $|l'| < 1$, and it does not rule out weaker evolution notions that do not furnish classical solutions for every datum in $D(A(s))$.

6 Verification notes

The included verification script checks $\text{rank } W_B = 2$, the identity $W_B \Sigma W_B^T = 0$, the two characteristic footpoints, and the trace contradiction at four small times. These computations only check the bookkeeping; equations (4)–(6) are the proof.

7 Novelty search and limitations

The lightweight run indexes were searched for arXiv:2302.03344 and the core keywords. Bounded web searches on 22 July 2026 used the exact title together with “moving interface”, “evolution family”, “characteristic speed”, “subcharacteristic”, “supersonic”, and “port-Hamiltonian”. They also checked nearby records arXiv:2301.07344, arXiv:2210.00056, and arXiv:2501.14930. No prior statement of this explicit obstruction or datum was located. This is a bounded novelty check, not a claim of exhaustive priority.

The recommended human-review focus is the solution concept: the conclusion is specifically about an evolution family giving classical solutions for all natural-domain data. The characteristic contradiction itself is independent of abstract generation machinery.

References

- [1] A. Kilian, B. Maschke, A. Mironchenko, and F. Wirth, *A case study of port-Hamiltonian systems with a moving interface*, arXiv:2302.03344; IEEE Control Systems Letters 7 (2023), 1572–1577.